Chapter 7. Hypothesis Testing

Logic of tests, likelihood ratio tests, large-sample approximations

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7.1 The logic of testing

A model is proposed on the basis of experience; we then ask whether the data agree with it. If, under the model, the observed result would be very unlikely, we reject the model in favour of an alternative. Deciding between a hypothesis and its alternative from data in this way is **statistical hypothesis testing**.

Example 7.1

The time until a standard painkiller takes effect is $N(20, 4^2)$ minutes. A new drug is claimed to act faster. In a trial with $n = 64$ patients the mean time is $\bar{X}$, and $\bar{X} \sim N(\mu, 4^2/64) = N(\mu, 0.5^2)$. The question is whether to accept $H_1 : \mu < 20$ (faster) over $H_0 : \mu = 20$ (no change).
Results with $\bar{X}$ well below 20 are hard to explain under H_0. For instance,

$$P(\bar{X} \le 19 \mid H_0) = P\Big(\frac{\bar{X} - 20}{0.5} \le -2\Big) = 0.023 :$$

such a result happens about twice in a hundred trials if H_0 holds. Observing it is strong evidence against H_0.

H_0 is the **null hypothesis** and H_1 the **alternative**. A set C of sample values for which we reject H_0 is the **rejection** (critical) region, and its complement is the **acceptance region**. A statistic used to describe C, such as $\bar{X}$ above, is a **test statistic**. Two kinds of error are possible:

	H_0 true	H_1 true
accept H_0	correct	type II error
reject H_0	type I error	correct

Since H_0 is the hypothesis we reject only on clear evidence, we bound the probability of a type I error by a small preassigned number α, the **significance level** (typically 0.10, 0.05, 0.01). A test with $P(\text{reject} \mid H_0) \le \alpha$ is a **level-α test**. Among level-α tests we want the smallest probability of a type II error, so we make the rejection region as large as the level permits.

Example 7.2

In Example 7.1, both "$\bar{X} \le 19$" (probability 0.023 under H_0) and "$\bar{X} \le 20 - z_{0.05} \cdot 0.5 = 19.1775$" (probability exactly 0.05) are level-0.05 tests. If the true mean is μ,

$$P_\mu(\bar{X} \le 19) = \Phi(38 - 2\mu), \qquad P_\mu(\bar{X} \le 19.1775) = \Phi(38.355 - 2\mu).$$

The second is larger for every $\mu < 20$, i.e. its type II error probability is smaller. In general the level-α rejection region here is $\bar{X} \le 20 - z_\alpha \cdot 0.5$.

Example 7.3 (Randomization)

Let $X_1, \ldots, X_{50}$ be a Poisson(θ) sample, and test $H_0 : \theta = 0.1$ against $H_1 : \theta < 0.1$ at $\alpha = 0.05$. Small values of $S = \sum X_i \sim \text{Poisson}(50\theta)$ speak against H_0, and under H_0, $S \sim \text{Poisson}(5)$:

$$P_{0.1}(S \le 0) = 0.0067, \qquad P_{0.1}(S \le 1) = 0.0404, \qquad P_{0.1}(S \le 2) = 0.1247.$$

No region of the form $S \le k$ has probability exactly 0.05. We can reach it by rejecting for $S \le 1$, rejecting with probability γ when $S = 2$, and accepting for $S \ge 3$, where

$$0.0404 + \gamma \cdot P_{0.1}(S = 2) = 0.05, \qquad \gamma = \frac{0.05 - 0.0404}{0.0842} = 0.114.$$

This is a **randomized test**.

General formulation. Let the model be $\{f(x;\theta):\theta\in\Omega\}$ and test

$$H_0:\theta\in\Omega_0 \quad \text{vs} \quad H_1:\theta\in\Omega_1, \qquad \Omega_0\cap\Omega_1=\varnothing,\ \Omega_0\cup\Omega_1=\Omega.$$

A (possibly randomized) test is described by a **test function** $\phi(\mathbf{x})\in[0,1]$, the probability of rejecting H_0 when $\mathbf{X}=\mathbf{x}$; a nonrandomized test has $\phi=\mathbf{1}_C$. Its **power function** is $\gamma_\phi(\theta)=E_\theta\phi(\mathbf{X})$, the probability of rejection under θ. Its **size** is $\max_{\theta\in\Omega_0}E_\theta\phi(\mathbf{X})$, and ϕ has level α if its size is at most α. We seek tests with size exactly α whose power on Ω_1 is as large as possible.

Example 7.4 (Composite null)

In Example 7.1 take $H_0:\mu\ge 20$ against $H_1:\mu<20$. For a region $\bar X\le c$, $P_\mu(\bar X\le c)=\Phi((c-\mu)/0.5)$ decreases in μ, so its maximum over $\mu\ge 20$ is attained at $\mu=20$. Setting it equal to α gives the same region $\bar X\le 20-0.5z_\alpha$ as before. Its power function $\gamma(\mu)=\Phi(38.355-2\mu)$ (for $\alpha=0.05$) decreases from 1 to 0, passing through 0.05 at $\mu=20$. In Example 7.3 the power function is

$$\gamma(\theta)=\sum_{x=0}^{1}e^{-50\theta}\frac{(50\theta)^x}{x!}+0.114\,e^{-50\theta}\frac{(50\theta)^2}{2},$$

which is also decreasing in θ.

7.2 Likelihood ratio tests

To test $H_0:\theta\in\Omega_0$ against $H_1:\theta\in\Omega_1$, compare how well the best parameter value in each hypothesis explains the data:

$$\frac{\max_{\theta\in\Omega_1}L(\theta;\mathbf{x})}{\max_{\theta\in\Omega_0}L(\theta;\mathbf{x})}.$$

Large values are evidence against H_0. Since $\max_\Omega L=\max\{\max_{\Omega_0}L,\max_{\Omega_1}L\}$, we have

$$\frac{\max_\Omega L}{\max_{\Omega_0}L}=\max\left\{1,\frac{\max_{\Omega_1}L}{\max_{\Omega_0}L}\right\},$$

and large values of one ratio correspond to large values of the other. So we use the full maximum. With $\hat\theta^{\Omega}$ and $\hat\theta^{\Omega_0}$ the MLEs over Ω and Ω_0:

Definition (Likelihood ratio test)

The **likelihood ratio test** (LRT) of level α rejects H_0 when

$$2\{l(\hat\theta^{\Omega})-l(\hat\theta^{\Omega_0})\}\ge c, \qquad \text{with } c \text{ chosen so that} \quad \max_{\theta\in\Omega_0}P_\theta(\text{reject})=\alpha$$

(randomizing if necessary).

Example 7.5 (Two-sided t test)

Let $X_1,\dots,X_n$ $(n\ge 2)$ be $N(\mu,\sigma^2)$ and test $H_0:\mu=\mu_0$ vs $H_1:\mu\ne\mu_0$. With

$$l(\mu,\sigma^2)=-\frac{1}{2\sigma^2}\sum(x_i-\mu)^2-\frac{n}{2}\log\sigma^2-\frac{n}{2}\log 2\pi,$$

the MLEs are $\hat\theta^{\Omega} = (\bar x, \hat\sigma^2)$, $\hat\sigma^2 = \frac{1}{n}\sum(x_i - \bar x)^2$, and $\hat\theta^{\Omega_0} = (\mu_0, \hat\sigma_0^2)$, $\hat\sigma_0^2 = \frac{1}{n}\sum(x_i - \mu_0)^2 = \hat\sigma^2 + (\bar x - \mu_0)^2$. Hence

$$2\{l(\hat\theta^{\Omega}) - l(\hat\theta^{\Omega_0})\} = n\log\frac{\hat\sigma_0^2}{\hat\sigma^2} = n\log\Big(1 + \frac{(\bar x - \mu_0)^2}{\hat\sigma^2}\Big),$$

an increasing function of $|\bar x - \mu_0|/(s/\sqrt{n})$ with $s^2 = \frac{1}{n-1}\sum(x_i - \bar x)^2$. Under H_0, $\sqrt{n}(\bar X - \mu_0)/S \sim t(n-1)$, so the level-$\alpha$ LRT is the **two-sided t test**:

$$\text{reject } H_0 \iff \Big|\frac{\bar X - \mu_0}{S/\sqrt{n}}\Big| \ge t_{\alpha/2}(n-1).$$

Example 7.6 (One-sided t test)

Same model, $H_0 : \mu \le \mu_0$ vs $H_1 : \mu > \mu_0$. Over Ω_0, $\sum(x_i - \mu)^2 = \sum(x_i - \bar x)^2 + n(\bar x - \mu)^2$ is minimized by $\hat\mu^{\Omega_0} = \min(\bar x, \mu_0)$, and $\hat\sigma^{2,\Omega_0} = \frac{1}{n}\sum(x_i - \hat\mu^{\Omega_0})^2$. Thus

$$2\{l(\hat\theta^{\Omega}) - l(\hat\theta^{\Omega_0})\} = n\log\Big(1 + \frac{(\bar x - \min(\bar x, \mu_0))^2}{\hat\sigma^2}\Big),$$

which is 0 when $\bar x \le \mu_0$ and increases in $(\bar x - \mu_0)/(s/\sqrt{n})$ when $\bar x > \mu_0$. For $\alpha < 1$ the region is $\{\sqrt{n}(\bar x - \mu_0)/s \ge c\}$ with $c > 0$. To find the size, write $Z = \sqrt{n}(\bar X - \mu)/\sigma$ and $V = (n-1)S^2/\sigma^2$, which are independent $N(0,1)$ and $\chi^2(n-1)$. Then

$$P_{\mu,\sigma^2}\Big(\frac{\bar X - \mu_0}{S/\sqrt{n}} \ge c\Big) = P\Big(c\sqrt{V/(n-1)} - Z \le \frac{\mu - \mu_0}{\sigma/\sqrt{n}}\Big),$$

which increases in μ. Over Ω_0 the maximum is at $\mu = \mu_0$, where the statistic is $t(n-1)$. The level-α LRT is the **one-sided t test**: reject when $\sqrt{n}(\bar X - \mu_0)/S \ge t_\alpha(n-1)$.

Example 7.7 (Exponential mean)

For an $\mathrm{Exp}(\theta)$ sample and $H_0 : \theta = \theta_0$ vs $H_1 : \theta \ne \theta_0$: $l(\theta) = -n\log\theta - n\bar x/\theta$, $\hat\theta^{\Omega} = \bar x$, so

$$2\{l(\hat\theta^{\Omega}) - l(\theta_0)\} = 2n\Big(\frac{\bar x}{\theta_0} - 1 - \log\frac{\bar x}{\theta_0}\Big).$$

The function $t \mapsto t - 1 - \log t$ decreases on $(0,1)$ and increases on $(1,\infty)$, so the region has the form

$$\frac{\bar x}{\theta_0} \le c_1 \quad\text{or}\quad \frac{\bar x}{\theta_0} \ge c_2, \qquad c_1 - \log c_1 = c_2 - \log c_2.$$

Under H_0, $2n\bar X/\theta_0 \sim \chi^2(2n)$, so the size condition is $\int_{2nc_1}^{2nc_2} f_{\chi^2(2n)}(y)\,dy = 1 - \alpha$. Solving the two equations gives c_1, c_2. For large n, $c_{1,2} \approx 1 \mp z_{\alpha/2}/\sqrt{n}$, and the test is approximately "$n(\bar X/\theta_0 - 1)^2 \ge \chi^2_\alpha(1)$".

Example 7.8 (Two-sample t test)

Independent samples $X_{i1}, \dots, X_{in_i}$ from $N(\mu_i, \sigma^2)$, $i = 1, 2$; $H_0 : \mu_1 = \mu_2$ vs $H_1 : \mu_1 \ne \mu_2$. Under Ω, $\hat\mu_i = \bar x_i$ and $\hat\sigma^2 = \sum_{i,j}(x_{ij} - \bar x_i)^2/(n_1 + n_2)$; under Ω_0, the common mean is $\hat\mu =$

$(n_1\bar{x}_1+n_2\bar{x}_2)/(n_1+n_2)$ and $\hat{\sigma}_0^2=\sum_{i,j}(x_{ij}-\hat{\mu})^2/(n_1+n_2)$. Since $\hat{\sigma}_0^2=\hat{\sigma}^2+\frac{n_1n_2}{(n_1+n_2)^2}(\bar{x}_1-\bar{x}_2)^2$,

$$2\{l(\hat{\theta}^{\Omega})-l(\hat{\theta}^{\Omega_0})\}=(n_1+n_2)\log\Big(1+\frac{n_1n_2}{(n_1+n_2)^2}\frac{(\bar{x}_1-\bar{x}_2)^2}{\hat{\sigma}^2}\Big),$$

increasing in $|\bar{x}_1-\bar{x}_2|/(s_p\sqrt{1/n_1+1/n_2})$, where $s_p^2=\sum_{i,j}(x_{ij}-\bar{x}_i)^2/(n_1+n_2-2)$. Under H_0 this statistic is $t(n_1+n_2-2)$ (Corollary 4.10 in Volume I). So the LRT rejects when

$$\frac{|\bar{X}_1-\bar{X}_2|}{S_p\sqrt{1/n_1+1/n_2}}\geq t_{\alpha/2}(n_1+n_2-2).$$

Example 7.9 (Comparing two variances)

Independent samples from $N(\mu_i,\sigma_i^2)$; $H_0:\sigma_1^2=\sigma_2^2$ vs $H_1:\sigma_1^2\neq\sigma_2^2$. With $\hat{\sigma}_i^2=\frac{1}{n_i}\sum_j(x_{ij}-\bar{x}_i)^2$ and the pooled $\hat{\sigma}^2=(n_1\hat{\sigma}_1^2+n_2\hat{\sigma}_2^2)/(n_1+n_2)$,

$$2\{l(\hat{\theta}^{\Omega})-l(\hat{\theta}^{\Omega_0})\}=(n_1+n_2)\log\hat{\sigma}^2-n_1\log\hat{\sigma}_1^2-n_2\log\hat{\sigma}_2^2.$$

In terms of $t=n_1\hat{\sigma}_1^2/(n_1\hat{\sigma}_1^2+n_2\hat{\sigma}_2^2)$ this equals $-n_1\log t-n_2\log(1-t)+\text{const}$, which decreases for $t<n_1/(n_1+n_2)$ and increases afterwards. Since t is an increasing function of $F=S_1^2/S_2^2$, the region is $F\leq d_1$ or $F\geq d_2$. Under H_0, $F\sim F(n_1-1,n_2-1)$, and d_1,d_2 are determined by $\int_{d_1}^{d_2}f_{F(r_1,r_2)}=1-\alpha$ together with equality of the LRT statistic at d_1 and d_2. In practice one uses $d_1=F_{1-\alpha/2}$, $d_2=F_{\alpha/2}$.

Example 7.10 (Poisson, one-sided)

In the setting of Example 7.3, $H_0:\theta\geq 0.1$ vs $H_1:\theta<0.1$. Here $\hat{\theta}^{\Omega}=\bar{x}$ and $\hat{\theta}^{\Omega_0}=\max(\bar{x},0.1)$, and

$$2\{l(\hat{\theta}^{\Omega})-l(\hat{\theta}^{\Omega_0})\}=2n\{\bar{x}\log\bar{x}-\bar{x}(1+\log 0.1)+0.1\}\mathbf{1}\{\bar{x}<0.1\},$$

which decreases in $\bar{x}$ on $\bar{x}\leq 0.1$. So the LRT rejects for small $S=\sum X_i$. Since $P_\theta(S\leq k)=\int_{n\theta}^{\infty}z^ke^{-z}/k!\,dz$ decreases in θ, the size is attained at $\theta=0.1$, and the randomized test of Example 7.3 is the level-0.05 LRT.

7.3 Large-sample approximations

When exact null distributions are unavailable, we use the asymptotic theory of the MLE. Let $\Omega\subseteq\mathbb{R}^k$ be open, $l(\theta)=\sum\log f(X_i;\theta)$, $\bar{l}=l/n$, and $I(\theta)$ the Fisher information (matrix). From Chapter 6, $\sqrt{n}(\hat{\theta}_n-\theta)\approx I(\theta)^{-1}\sqrt{n}\,\dot{\bar{l}}(\theta)$.

Simple null hypothesis. Let $H_0:\theta=\theta_0$. Expanding l about $\hat{\theta}_n$, where $\dot{l}(\hat{\theta}_n)=0$,

$$l(\theta_0)=l(\hat{\theta}_n)+\tfrac{1}{2}(\theta_0-\hat{\theta}_n)'\ddot{l}(\theta_n^*)(\theta_0-\hat{\theta}_n),$$

and $-\frac{1}{n}\ddot{l}(\theta_n^*)\xrightarrow{p}I(\theta_0)$ under H_0. Hence

$$2\{l(\hat{\theta}_n)-l(\theta_0)\}\approx\sqrt{n}(\hat{\theta}_n-\theta_0)'I(\theta_0)\sqrt{n}(\hat{\theta}_n-\theta_0).$$

Since $\sqrt{n}(\hat{\theta}_n-\theta_0)\xrightarrow{d}\mathbf{Z}\sim N_k(\mathbf{0},I(\theta_0)^{-1})$, the right side converges to $\mathbf{Z}'I(\theta_0)\mathbf{Z}\sim\chi^2(k)$ (continuous mapping, and Theorem 4.18 in Volume I).

Theorem 7.1 (Simple null hypothesis)

Under the conditions (R0)–(R7) of Theorem 6.13, for $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$, the following hold when H_0 is true.

(a) $2\{l(\hat\theta_n) - l(\theta_0)\} \xrightarrow{d} \chi^2(k)$. So the test rejecting when this is $\geq \chi^2_\alpha(k)$ has asymptotic level α.

(b) $2\{l(\hat\theta_n) - l(\theta_0)\} = W_n(\theta_0) + o_p(1)$, where $W_n(\theta_0) = n(\hat\theta_n - \theta_0)' I(\theta_0)(\hat\theta_n - \theta_0)$.

(c) $2\{l(\hat\theta_n) - l(\theta_0)\} = R_n(\theta_0) + o_p(1)$, where $R_n(\theta_0) = n\,\dot l(\theta_0)' I(\theta_0)^{-1} \dot l(\theta_0)$.

Sketch. (a), (b) were derived above; the remainder is controlled as in the proof of Theorem 6.13. For (c), substitute $\sqrt{n}(\hat\theta_n - \theta_0) = I(\theta_0)^{-1}\sqrt{n}\,\dot l(\theta_0) + o_p(1)$ into W_n. ∎

W_n is **Wald's statistic** and R_n is **Rao's (score) statistic**. All three are asymptotically $\chi^2(k)$ under H_0, and rejecting when any of them exceeds $\chi^2_\alpha(k)$ is an asymptotic level-α test. The score test needs only θ_0, not the MLE.

Example 7.11 (Exponential)

For $\mathrm{Exp}(\theta)$, $H_0 : \theta = \theta_0$: $\dot l(\theta) = -\frac{1}{\theta} + \frac{\bar x}{\theta^2}$, $\hat\theta = \bar x$, $I(\theta) = 1/\theta^2$. The three statistics are

$$2\{l(\hat\theta) - l(\theta_0)\} = 2n\Big(\frac{\bar x}{\theta_0} - 1 - \log\frac{\bar x}{\theta_0}\Big), \qquad W_n = \frac{n(\bar x - \theta_0)^2}{\bar x^2}, \qquad R_n = \frac{n(\bar x - \theta_0)^2}{\theta_0^2}.$$

Each is compared with $\chi^2_\alpha(1)$.

Example 7.12 (Multinomial goodness of fit)

Let $\mathbf{Z}_1, \dots, \mathbf{Z}_n$ be i.i.d. $\mathrm{Multi}(1, \mathbf{p})$ with counts $X_j = \sum_i Z_{ij}$, and test $H_0 : \mathbf{p} = \mathbf{p}_0$. Use $\theta = (p_1, \dots, p_r)'$, $r = k - 1$, and write $\theta_\cdot = \sum_{j \leq r} \theta_j = 1 - p_k$. Then

$$l(\theta) = \sum_{j \leq r} x_j \log\theta_j + x_k \log(1 - \theta_\cdot), \qquad \hat\theta_j = x_j/n,$$

$$I(\theta) = \mathrm{diag}(1/\theta_j) + \frac{1}{p_k}\mathbf{1}\mathbf{1}'.$$

A direct computation gives $I(\theta_0)(\hat\theta - \theta_0) = \dot l(\theta_0)$, with j-th entry $\frac{x_j - np_{0j}}{np_{0j}} - \frac{x_k - np_{0k}}{np_{0k}}$, and therefore

$$W_n(\theta_0) = R_n(\theta_0) = \sum_{j=1}^{k} \frac{(X_j - np_{0j})^2}{np_{0j}},$$

Pearson's chi-square statistic, while the LRT statistic is $2\sum_j X_j \log\frac{X_j}{np_{0j}}$. All are asymptotically $\chi^2(k-1)$ under H_0 (compare Example 5.12 in Volume I).

Composite null hypotheses. Often H_0 fixes only part of θ, e.g. the mean of a normal model with unknown variance. Write $\theta = (\xi', \eta')'$ with $\xi \in \mathbb{R}^{k_1}$, $\eta \in \mathbb{R}^{k_0}$, $k_0 + k_1 = k$, and $H_0 : \xi = \xi_0$. Partition

$$\dot l = \begin{pmatrix} \dot l_1 \\ \dot l_2 \end{pmatrix}, \qquad I = \begin{pmatrix} I_{11} & I_{12} \\ I_{21} & I_{22} \end{pmatrix}, \qquad I_{11\cdot 2} = I_{11} - I_{12} I_{22}^{-1} I_{21}.$$

Let $\hat\theta_n$ be the unrestricted MLE and $\hat\theta_n^0 = (\xi_0', \hat\eta_n^{0\prime})'$ the MLE under H_0.

Theorem 7.2 (Composite null hypothesis)

Under the conditions of Theorem 6.13, if $H_0 : \xi = \xi_0$ is true with true value $\theta_0 = (\xi_0', \eta_0')'$, then

(a) $2\{l(\hat\theta_n) - l(\hat\theta_n^0)\} \xrightarrow{d} \chi^2(k - k_0)$;

(b) $2\{l(\hat\theta_n) - l(\hat\theta_n^0)\} = n(\hat\theta_n - \hat\theta_n^0)' I(\theta_0)(\hat\theta_n - \hat\theta_n^0) + o_p(1)$;

(c) $2\{l(\hat\theta_n) - l(\hat\theta_n^0)\} = n\,\dot l(\hat\theta_n^0)' I(\theta_0)^{-1} \dot l(\hat\theta_n^0) + o_p(1)$.

Sketch. As in Theorem 7.1,

$$2\{l(\hat\theta_n) - l(\theta_0)\} \approx \sqrt{n}\,\dot l(\theta_0)' I^{-1} \sqrt{n}\,\dot l(\theta_0).$$

Applying the same argument to the submodel with $\xi = \xi_0$ fixed, where only η is estimated, gives

$$2\{l(\hat\theta_n^0) - l(\theta_0)\} \approx \sqrt{n}\,\dot l_2(\theta_0)' I_{22}^{-1} \sqrt{n}\,\dot l_2(\theta_0).$$

Subtracting, and using the formula for the inverse of a partitioned matrix,

$$\begin{pmatrix} I_{11} & I_{12} \\ I_{21} & I_{22} \end{pmatrix}^{-1} = \begin{pmatrix} I_{11\cdot 2}^{-1} & -I_{11\cdot 2}^{-1} I_{12} I_{22}^{-1} \\ -I_{22}^{-1} I_{21} I_{11\cdot 2}^{-1} & I_{22}^{-1} + I_{22}^{-1} I_{21} I_{11\cdot 2}^{-1} I_{12} I_{22}^{-1} \end{pmatrix},$$

one finds

$$2\{l(\hat\theta_n) - l(\hat\theta_n^0)\} \approx \sqrt{n}\,\mathbf{Z}_n' I_{11\cdot 2}^{-1} \sqrt{n}\,\mathbf{Z}_n, \qquad \mathbf{Z}_n = \dot l_1(\theta_0) - I_{12} I_{22}^{-1} \dot l_2(\theta_0).$$

By the CLT, $\sqrt{n}\,\mathbf{Z}_n \xrightarrow{d} N_{k_1}(\mathbf{0}, I_{11\cdot 2})$, so the limit is $\chi^2(k_1) = \chi^2(k - k_0)$. This proves (a). For (b) one checks $\hat\eta_n^0 - \eta_0 \approx I_{22}^{-1} \dot l_2(\theta_0)$ and $I_{11\cdot 2}(\hat\xi_n - \xi_0) \approx \mathbf{Z}_n$, so that $n(\hat\theta_n - \hat\theta_n^0)' I (\hat\theta_n - \hat\theta_n^0) \approx n \mathbf{Z}_n' I_{11\cdot 2}^{-1} \mathbf{Z}_n$; (c) is similar. ∎

The statistics in (b) and (c), with $I(\theta_0)$ replaced by $I(\hat\theta_n^0)$ (since η_0 is unknown), are the Wald and Rao statistics for composite hypotheses. A null hypothesis of the form $H_0 : g_1(\theta) = \cdots = g_{k_1}(\theta) = 0$ with smooth g_j is handled by reparametrizing so that $\xi_j = g_j(\theta)$; the limit is $\chi^2(k_1)$, where k_1 is the number of constraints, i.e. $\dim \Omega - \dim \Omega_0$.

Example 7.13 (Independence in a contingency table)

Counts X_{ij} ($i \le r$, $j \le c$) of an $r \times c$ table are Multi$(n, (p_{ij}))$. Independence is $H_0 : p_{ij} = p_{i\cdot} p_{\cdot j}$ for all i, j. The MLEs are $\hat p_{ij} = x_{ij}/n$ in general and $\hat p_{ij}^0 = \frac{x_{i\cdot}}{n}\frac{x_{\cdot j}}{n}$ under H_0, so

$$2\{l(\hat p) - l(\hat p^0)\} = 2 \sum_{i,j} X_{ij} \log \frac{X_{ij}}{n \hat p_{i\cdot} \hat p_{\cdot j}}.$$

The dimensions are $\dim \Omega = rc - 1$ and $\dim \Omega_0 = (r-1) + (c-1)$, so the limit is $\chi^2((r-1)(c-1))$. If the marginals were known, Example 7.12 would give Pearson's statistic with $np_{i\cdot}p_{\cdot j}$; substituting their estimates gives the usual **chi-square test of independence**:

$$\sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - \hat E_{ij})^2}{\hat E_{ij}} \ge \chi_\alpha^2((r-1)(c-1)), \qquad O_{ij} = X_{ij},\ \hat E_{ij} = \frac{X_{i\cdot} X_{\cdot j}}{n}.$$

Example 7.14 (A worked 2×3 table)

A survey of 200 students classified by year (lower, upper) and preferred study place (library, home, cafe) gives

	library	home	cafe	total
lower	40	30	30	100
upper	20	50	30	100
total	60	80	60	200

The expected counts are $\hat{E}_{1j} = \hat{E}_{2j} = (30, 40, 30)$. Pearson's statistic is

$$2\Big(\frac{10^2}{30} + \frac{10^2}{40} + 0\Big) = 11.67 > \chi^2_{0.05}(2) = 5.99,$$

so independence is rejected at the 5% level.

Non-identically distributed observations. The same results hold, under suitable conditions, when observations are independent but not identically distributed. Then $\bar{I}_n(\theta) = \frac{1}{n} E_\theta[-\ddot{l}(\theta)]$ replaces $I(\theta)$, and $W_n = n(\hat{\theta} - \theta_0)' \bar{I}_n(\theta_0)(\hat{\theta} - \theta_0)$, $R_n = n\,\dot{l}(\theta_0)' \bar{I}_n(\theta_0)^{-1} \dot{l}(\theta_0)$.

Example 7.15 (Homogeneity of multinomial populations)

Let $\mathbf{X}_i \sim \text{Multi}(n_i, \mathbf{p}_i)$, $i = 1, \dots, r$, be independent, each with c cells, and test $H_0 : \mathbf{p}_1 = \cdots = \mathbf{p}_r$. Under H_0 the common $\mathbf{p}$ is estimated by pooling, $\hat{p}^0_j = X_{\cdot j}/n$, $n = \sum n_i$. With $O_{ij} = X_{ij}$ and $\hat{E}_{ij} = n_i \hat{p}^0_j$, the LRT statistic is $2 \sum O_{ij} \log(O_{ij}/\hat{E}_{ij})$. Expanding $2u \log u \approx 2(u-1) + (u-1)^2$ around $u = O_{ij}/\hat{E}_{ij} = 1$, it is approximately Pearson's $\sum \sum (O_{ij} - \hat{E}_{ij})^2 / \hat{E}_{ij}$. The degrees of freedom are $r(c-1) - (c-1) = (r-1)(c-1)$, the same as for independence, although the sampling scheme differs: here the row totals are fixed. (One shows the limit by writing the statistic as a quadratic form in the asymptotically normal vectors $\sqrt{n_i}(\hat{\mathbf{p}}_i - \hat{\mathbf{p}}^0)$ with an idempotent matrix of rank $(r-1)(c-1)$.)

7.4 Selected exercises

Motivation. These are textbook exercises for this chapter (numbered as in the book). They derive likelihood ratio tests in models not treated above, show that the power of the t test grows with the standardized effect, and explain why the Poisson homogeneity statistic is approximately Pearson's χ^2.

Exercise 7.6 (Power of the two-sided t test)

Let $X_1, \dots, X_n$ $(n \geq 2)$ be $N(\mu, \sigma^2)$ and consider the two-sided t test of Example 7.5 with statistic $t = (\bar{X} - \mu_0)/(S/\sqrt{n})$.

(a) Show that $t \overset{d}{=} (Z + \delta)/\sqrt{V/(n-1)}$, where $\delta = (\mu - \mu_0)/(\sigma/\sqrt{n})$, $Z \sim N(0,1)$, $V \sim \chi^2(n-1)$, and Z, V are independent.

(b) Show that the power $P_{\mu,\sigma^2}\{|t| \geq t_{\alpha/2}(n-1)\}$ is a function of $|\delta|$ alone and is increasing in $|\delta|$.

Solution.

(a) Put $Z = (\bar{X} - \mu)/(\sigma/\sqrt{n})$ and $V = (n-1)S^2/\sigma^2$. For a normal sample $\bar{X}$ and S^2 are independent, $Z \sim N(0,1)$ and $V \sim \chi^2(n-1)$ (the normal-sample theorem of Chapter 4). Dividing numerator

and denominator of t by $\sigma/\sqrt{n}$,

$$t = \frac{(\bar{X}-\mu)/(\sigma/\sqrt{n}) + (\mu-\mu_0)/(\sigma/\sqrt{n})}{S/\sigma} = \frac{Z+\delta}{\sqrt{V/(n-1)}}.$$

(b)

Step 1. Let $c = t_{\alpha/2}(n-1)$ and, for $k>0$, $g_k(\delta) = P(|Z+\delta| \ge k) = 1 - \Phi(k-\delta) + \Phi(-k-\delta)$. By (a) and independence of Z and V,

$$\gamma(\delta) = P(|t| \ge c) = E\big[g_K(\delta)\big], \qquad K = c\sqrt{V/(n-1)}.$$

Step 2. Since $-Z \overset{d}{=} Z$, $g_k(-\delta) = P(|-Z+\delta| \ge k) = g_k(\delta)$, so $\gamma(-\delta) = \gamma(\delta)$: the power depends on $|\delta|$ only.

Step 3. $g_k'(\delta) = \phi(k-\delta) - \phi(k+\delta)$. For $\delta > 0$, $|k-\delta| < k+\delta$, hence $\phi(k-\delta) > \phi(k+\delta)$ and g_k is strictly increasing on $[0,\infty)$ for every $k>0$.

Step 4. $K>0$ with probability 1, so for $0 \le \delta_1 < \delta_2$ we have $g_K(\delta_1) < g_K(\delta_2)$ pointwise, and taking expectations gives $\gamma(\delta_1) < \gamma(\delta_2)$. The law of t is the *noncentral t distribution* with noncentrality δ. ■

Exercise 7.9 (A one-sided test for Beta$(\theta,1)$)

Let $X_1,\dots,X_n$ have density $f(x;\theta) = \theta x^{\theta-1}$ on $(0,1)$, $\theta>0$. Find the level-α LRT of $H_0:\theta \le 1$ against $H_1:\theta>1$ $(0<\alpha<\frac{1}{2})$.

Solution.

Step 1. With $T = -\sum_i \log X_i > 0$, the log-likelihood is $l(\theta) = n\log\theta - (\theta-1)T$, strictly concave with maximum at $\hat\theta = n/T$. The MLE over $\Omega_0 = (0,1]$ is $\min(\hat\theta, 1)$ by concavity.

Step 2. If $\hat\theta \le 1$ the two maxima agree and the statistic is 0. If $\hat\theta > 1$, i.e. $T<n$,

$$2\{l(\hat\theta) - l(1)\} = 2\{n\log(n/T) - n + T\} =: h(T), \qquad h'(T) = 2\Big(1 - \frac{n}{T}\Big) < 0.$$

So h decreases from $+\infty$ to 0 on $(0,n)$, and for every $c>0$ the event $\{\text{statistic} \ge c\}$ equals $\{T \le d\}$ for some $d \in (0,n)$. The LRT rejects H_0 for small T.

Step 3. Under θ, $P(-\log X_i > s) = P(X_i < e^{-s}) = e^{-\theta s}$, so the $-\log X_i$ are i.i.d. exponential with rate θ and $2\theta T \sim \chi^2(2n)$. Hence $P_\theta(T \le d) = P\{\chi^2(2n) \le 2\theta d\}$ increases with θ, and the size is attained at $\theta = 1$.

Step 4. The level-α LRT therefore rejects H_0 iff

$$-2\sum_{i=1}^{n} \log X_i \le \chi^2_{1-\alpha}(2n),$$

the lower α point of $\chi^2(2n)$. Since $\alpha < \frac{1}{2}$, $\chi^2_{1-\alpha}(2n)$ is below the median of $\chi^2(2n)$, which is known to lie below its mean $2n$ (the median of a gamma law is smaller than its mean); so $d = \chi^2_{1-\alpha}(2n)/2 < n$ as required in Step 2. ■

Exercise 7.15(a) (Testing the endpoint of a uniform law)

Let $X_1,\dots,X_n$ be $U[0,\theta]$, $\theta>0$. Find the level-α LRT of $H_0:\theta=\theta_0$ against $H_1:\theta\ne\theta_0$.

Solution.

Step 1. With $M = \max_i X_i$, $L(\theta) = \theta^{-n} I(\theta \geq M)$, so the MLE is $\hat{\theta} = M$ and

$$\frac{L(\theta_0)}{L(\hat{\theta})} = \begin{cases} (M/\theta_0)^n, & M \leq \theta_0, \\ 0, & M > \theta_0. \end{cases}$$

The LRT rejects when this ratio is small, i.e. when $M > \theta_0$ or $M \leq k\theta_0$ for a constant $k \in (0,1)$.

Step 2. Under H_0, $P(M > \theta_0) = 0$ and $P(M \leq k\theta_0) = k^n$. Setting $k^n = \alpha$ gives the level-α LRT

$$\text{reject } H_0 \iff M > \theta_0 \text{ or } M \leq \alpha^{1/n}\theta_0,$$

whose size is exactly α. Unlike the regular cases of Section 7.3, the statistic $2\{l(\hat{\theta}) - l(\theta_0)\} = -2n\log(M/\theta_0)$ is exactly $\chi^2(2)$ under H_0, not $\chi^2(1)$: $-n\log(M/\theta_0)$ is standard exponential because $P(M/\theta_0 \leq e^{-s/n}) = e^{-s}$. ■

Exercise 7.17(a)(b) (Homogeneity of Poisson means)

Let $X_{i1}, \ldots, X_{in_i}$ be a sample from Poisson(θ_i), $i = 1, \ldots, k$ $(k \geq 2)$, the samples being independent. Put $X_{i\cdot} = \sum_j X_{ij}$, $n = n_1 + \cdots + n_k$, $\hat{\theta}_i = X_{i\cdot}/n_i$ and $\hat{\theta}^0 = \sum_i X_{i\cdot}/n$. For $H_0 : \theta_1 = \cdots = \theta_k$:
(a) Show that the LRT statistic is $2\sum_i n_i\{\hat{\theta}_i \log(\hat{\theta}_i/\hat{\theta}^0) - \hat{\theta}_i + \hat{\theta}^0\}$ (with $0\log 0 = 0$).
(b) Show that if $n_i \to \infty$ with $n_i/n \to \gamma_i \in (0,1)$, then under H_0 (with common value $\theta > 0$) the statistic is approximately $\sum_i (X_{i\cdot} - n_i\hat{\theta}^0)^2/(n_i\hat{\theta}^0)$, in the sense that the ratio of the two tends to 1 in probability.

Solution.

(a) Up to a term free of θ, $l(\theta) = \sum_i \{X_{i\cdot}\log\theta_i - n_i\theta_i\}$. Each summand is maximized at $\hat{\theta}_i$, and under H_0 the function $(\sum_i X_{i\cdot})\log\vartheta - n\vartheta$ is maximized at $\hat{\theta}^0$. Using $X_{i\cdot} = n_i\hat{\theta}_i$,

$$2\{l(\hat{\theta}) - l(\hat{\theta}^0)\} = 2\sum_i \{n_i\hat{\theta}_i \log(\hat{\theta}_i/\hat{\theta}^0) - n_i\hat{\theta}_i + n_i\hat{\theta}^0\}.$$

(The linear terms sum to 0, since $\sum_i n_i\hat{\theta}_i = n\hat{\theta}^0$.)

(b)

Step 1. For $a > 0$ let $g_a(x) = x\log(x/a) - x + a$, $x \geq 0$. Then $g_a(a) = 0$, $g_a'(a) = \log(x/a)|_{x=a} = 0$ and $g_a''(x) = 1/x$. By Taylor's theorem with the Lagrange remainder, for $x > 0$

$$g_a(x) = \frac{(x-a)^2}{2\xi}, \qquad \xi \text{ between } x \text{ and } a.$$

Step 2. Under H_0, the law of large numbers gives $\hat{\theta}_i \xrightarrow{p} \theta$ and $\hat{\theta}^0 \xrightarrow{p} \theta$, so $P(\hat{\theta}_i > 0) \to 1$ and the intermediate points satisfy $\xi_i \xrightarrow{p} \theta$. Hence

$$2\{l(\hat{\theta}) - l(\hat{\theta}^0)\} = \sum_i n_i \frac{(\hat{\theta}_i - \hat{\theta}^0)^2}{\hat{\theta}^0} \cdot \frac{\hat{\theta}^0}{\xi_i},$$

and each factor $\hat{\theta}^0/\xi_i \xrightarrow{p} 1$.

Step 3. A sum $\sum_i a_i b_i$ with $a_i \geq 0$, $\sum_i a_i > 0$ lies between $\min_i b_i \sum_i a_i$ and $\max_i b_i \sum_i a_i$; with k fixed and every $b_i \xrightarrow{p} 1$, the ratio $\sum_i a_i b_i / \sum_i a_i \xrightarrow{p} 1$. Since $n_i(\hat{\theta}_i - \hat{\theta}^0)^2/\hat{\theta}^0 = (X_{i\cdot} - n_i\hat{\theta}^0)^2/(n_i\hat{\theta}^0)$, the statistic is the Pearson-type sum times factors tending to 1, which is the claim. (Its limit law is $\chi^2(k-1)$ by Theorem 7.2.) ■